

AI Basics

What is Artificial Intelligence

Artificial Intelligence, or AI, refers to computer systems that perform tasks which typically require human intelligence, such as understanding language, recognizing images, or making decisions. Modern AI systems are largely built using machine learning, a subfield in which models learn patterns from data rather than following explicitly programmed rules.

Machine Learning Fundamentals

- Supervised learning trains a model on labeled examples to predict outcomes for new data.
- Unsupervised learning finds patterns or groupings in data without labeled outcomes.
- Reinforcement learning trains an agent to make decisions through trial and error using rewards.

Most machine learning workflows involve collecting and cleaning data, selecting or engineering features, training a model, and evaluating its performance against a held-out test set before it is deployed.

Neural Networks

A neural network is composed of layers of interconnected nodes, loosely inspired by biological neurons. Each connection has a weight that is adjusted during training through a process called backpropagation, which uses gradient descent to reduce the difference between the model's predictions and the true values.

Large Language Models

Large language models, or LLMs, are neural networks trained on massive amounts of text to predict the next word in a sequence. Through this simple training objective at large scale, they develop the ability to summarize text, answer questions, translate languages, and generate code. Most modern LLMs are based on the transformer architecture, which uses a mechanism called attention to weigh the relevance of different words in a sequence.

Retrieval-Augmented Generation

Retrieval-Augmented Generation, or RAG, is a technique that combines a language model with an external knowledge source. Instead of relying only on what the model learned during training, relevant documents are retrieved from a database, typically using vector similarity search, and provided to the model as additional context before it generates a response. This approach reduces hallucination and allows the system to answer questions about information the model was never trained on, such as internal company documents.

Vector Embeddings

An embedding is a numerical representation of text, images, or other data as a vector in a high-dimensional space, where semantically similar items are placed close together. Embeddings are the foundation of semantic search: a query is converted into a vector and compared against stored document vectors using a similarity metric such as cosine similarity to find the most relevant matches.

Responsible AI Considerations

- Bias: models can reflect or amplify biases present in their training data.
- Privacy: systems that process personal data must handle and mask sensitive information carefully.
- Transparency: users benefit from understanding when they are interacting with an AI system.
- Reliability: outputs should be evaluated and monitored, since models can produce incorrect answers confidently.